EXOTIC SMOOTH STRUCTURE ON THE 7-SPHERE

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Abstract. In this article we construct and show that there are manifolds homeomorphic, but not diffeomorphic to the standard 7-sphere. We start with the basic definitions of fiber bundles and present two classification theorems. We then introduce the characteristic classes, leading to the statement of the Hirzebruch signature theorem. We apply the knowledge in these two sections to the familiar quaternion Hopf fibration to identify which element it represents in the classification of sphere bundles, and to compute its first Pontryagin class. Then we turn to the actual construction of the exotic spheres. We mainly follow Milnor's argument in [Mil56] with extra details and clarifications. We prioritize the quaternion Hopf fibration as the main example of the series of manifolds Milnor constructs.

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1. Introduction

In this article, we will discuss the construction of some exotic 7-spheres, that is, spaces that are homeomorphic to S^7 but whose differential structure is not compatible with the standard differential structure. The spaces will be built up from simple spaces — they will be constructed as S^3 bundles over S^4 with some peculiar twisting. The fiber bundle structure gives rise to a natural smooth structure. We will study this new smooth structure via the characteristic classes of the tangent bundle it induces. These are invariants of vector bundles with various properties we can take advantage of. Finally, a Morse theory argument will be applied to show that these twisted spaces are, in fact, homeomorphic to the 7-sphere.

We assume knowledge in basic smooth manifold theory, basic algebraic topology (equivalent to the main contents of [Hat22]), and some basic theory of vector bundles (sections, isomorphism of vector bundles, and pull-back bundles). We will cite [Loo26] and [Hat22] for results that totally belong to these areas.

2. Fiber bundle with structure group

Our journey begins with the notion of fiber bundle with structure group, due to Steenrod [Ste99, Definition 2.3].

Date: March 12, 2026.

Definition 2.1. A coordinate bundle ξ is a collection as follows:

(1) A triple of spaces (E, B, F) called the total space, the base space, and the fiber space,
(2) an onto map $p : E \to B$ called the projection,
(3) a topological group G acting effectively on F called the structure group of the bundle,
(4) and an open cover $\{V_j\}_{j\in J}$ of B, called coordinate neighborhoods, such that for any j in J, there exists a homeomorphism $\phi_j : V_j \times F \to p^{-1}(V_j)$, called the coordinate function.

The coordinate functions are required to satisfy the following conditions:

(5) For any $x \in V_j$, $y \in F$, $p\,\phi_j(x, y) = x$;
(6) If the map $\phi_{j,x} : F \to p^{-1}(x)$ is defined as $\phi_{x,j}(y) = \phi_j(x, y)$, then for each pair $i, j \in J$, and each $x \in V_i \cap V_j$, the homeomorphism, called transition functions.
$$\phi_{j,x}^{-1}\phi_{i,x} : F \to F$$
coincide with an element of G, and
(7) for each pair $i, j \in J$, the map $g_{ji} : V_i \cap V_j \to G$ defined by $g_{ji}(x) = \phi_{j,x}^{-1}\phi_{i,x}$ is continuous.

Remark 2.2. A group action is called *effective* if no element other than the identity acts trivially. Thus the structure group G is isomorphic to a group of homeomorphism of the fiber space. This prevents simply taking a gigantic group and giving it a very fine topology.

Two coordinate bundles with the same total space, base space, projection, fiber space, and structure group are said to be *equivalent in the strict sense* if the union of the open covers with their coordinate functions still define such a collection. It can be easily verified that this is an equivalence relation. A fiber bundle will be defined as an equivalence class of coordinate bundles. Sometimes we will write $F \to E \to B$ to indicate a fiber bundle with total space, base space, and fiber space equal to (E, B, F).

Remark 2.3. A rank n vector bundle is the special case where the fiber space F is an n-dimensional vector space, and the structure group is the general linear group $GL(n, \mathbb{R})$. Our first job is to extend the various notions of vector bundles to the case of fiber bundles. They are essentially similar, with some added requirements on the structure group, which is usually captured by the word 'linear' in the corresponding definitions for vector bundles.

Definition 2.4. Let $\mathcal{B}$ and $\mathcal{B}'$ be two coordinate bundles having the same fiber space and structure group. A continuous map $h : X \to X'$ is called a *bundle map* if

(1) h takes each fiber of F_x over B homeomorphically onto a fiber $F_{x'}$ over B';
(2) if $x \in V_j \cap h^{-1}(V_k')$, and $h_x : F_x \to F_{x'}$ is the restriction of h to F_x, then the map $\overline{g}_{kj}(x) := \phi_{k,x}^{'-1} h_x \phi_{j,x}$ coincide with an element in G, and
(3) the induced map $\overline{g}_{kj} : V_j \cap h^{-1}(V_k') \to G$ is continuous.

Definition 2.5. Two coordinate bundles having the same base space, fiber space, and structue group are said to be *equivalent* if there exists a bundle map between them that is the identity map on the base space.

Definition 2.6. A coordinate bundle $\mathcal{B}$ is called a trivial bundle if it is equivalent to a bundle where there is just one coordinate neighborhood.

Remark 2.7. The usual definition of fiber bundle (as in [Hat22], P377) does not involve the structure group G, which seemingly makes the definition unnecessarily longer. Indeed, for a fiber bundle defined as in [Hat22], we can just take G to be the automorphism group of the fiber space and give it the compact-open topology. However, this provides us with a clear way to construct a fiber bundle first on an open cover where the bundle over each open subset is trivial and then glue them together by specifying the transition function. This is made precise by the following theorem.

Let G be a topological group, and B a space. *A system of coordinate transformations in B with values in G* is an indexed open covering $\{V_j\}_{j\in J}$ of B and a collection of continuous maps $g_{ji} : V_i \cap V_j \to G$ for any $i, j \in J$ such that $g_{kj}(x)\, g_{ji}(x) = g_{ki}(x)$ for any $x \in V_i \cap V_j \cap V_k$.

Theorem 2.8. *If G is a topological group acting effectively on F, and $\{V_j\}$, $\{g_{ij}\}$ is a system of coordinate transformations in B, then there exists a fiber bundle with base space B, fiber space F, structure group G, and transition functions $\{g_{ij}\}$. Moreover, any such two bundles are equivalent.*

Proof. Form a big space $B \times F \times J$ with J having the discrete topology, and then mod out the desired identifications. For details see [Ste99, Theorem 3.2]. □

Note that in this theorem, the assumptions do not involve the fiber space but merely coordinate transformations on the base space. This indicates that the fiber space actually plays an auxiliary role, and that classifying bundles is equivalent to classifying coordinate transformations. This is made precise in the following theorem.

Theorem 2.9. *The operation of assigning to each bundle with base space, fiber space, structure group, and a system of coordinate transformations sets up a 1-1 correspondence between equivalence classes of bundles and equivalence classes of systems of coordinate transformations.*

Proof. See [Ste99, Lemma 2.8, Theorem 3.3]. □

One extra observation is that we are allowed to freely switch the fiber space, provided that the structure group G still acts on the space effectively. By choosing a nice fiber space, we may obtain stronger or better results. This is another advantage of introducing the structure group: in a sense, it is the more 'primitive' ingredient of a fiber bundle. This is the idea of associated principal bundle.

Definition 2.10. A coordinate bundle $\mathcal{B}$ is called a *principal bundle* if its fiber space is equal to its structure group G and G acts on it by left translations. For an arbitrary bundle, the *associated principal bundle* is the bundle given by the construction in Theorem 2.8 using the same base space, structure group, system of coordinate transformations, but replacing the fiber space by G and allowing the structure group to act on it by left translations.

To illustrate the the usefulness of considering the principal bundle, we state the following result.

Definition 2.11. A cross-section of a coordinate bundle is a continuous map $f : B \to E$ such that $pf(x) = x$ for any $x \in B$.

Theorem 2.12. *A principal bundle with group G is trivial if and only if it admits a cross-section.* [Ste99, Theorem 8.3]

Remark 2.13. Note that unlike vector bundles, fiber bundles don't have a 'zero-section'. In general, there is not a canonical way to select a base point. The reason why vector bundles do is that the origin of a vector space is invariant under any linear transformation so it is distinguished. In fact, the above theorem shows that for a principal bundle, possessing a cross-section would directly imply it is trivial. In contrast, a rank n vector bundle is trivial if and only if it possesses n cross-sections that are linearly independent at the fiber space of each point. [MS74, Theorem 2.2]

One of the primary objectives in bundle theory is to classify these bundles up to equivalence classes. In the following we introduce two classification theorems: one for bundles over spheres, and one for the general case. They turn out to be both very useful in later discussions.

For fiber bundles over spheres, they actually possess a simple structure. The sphere can be covered by two charts: the northern and southern hemisphere. If we restrict the fiber bundle to one of the chart, it must be trivial since the base space is contractible. Thus, fiber bundles over S^n are completely determined by one transition function on the 'equator' $(-\epsilon, \epsilon) \times S^{n-1}$, which is a

map from $(-\epsilon, \epsilon) \times S^{n-1}$ to the structure group G. This is a complete characterization of the fiber bundle. The next theorem states that, in fact, the equivalence class of the bundle only depends on the homotopy class of this map.

Theorem 2.14. *The equivalence classes of bundles over S^n with structure group G are in 1-1 correspondence with $\pi_{n-1}(G)$, provided that G is connected.* [Ste99, Theorem 18.5]

Now we turn to the general case. Here we need to use the additional structure of principal G-bundle. A principal G-bundle ω over B_0 is called *universal* if for any principal G-bundle ξ over a paracompact space B, there is a map $f : B \to B_0$ such that $\xi \cong f^*\omega$, and, furthermore, if f and g are two such maps, they are homotopic. It's clear that such a universal bundle, if exists, is unique for any group G. The base space B_0 is usually denoted as BG, and is called a classifying space for principal G-bundles. In other words, there is a bijection between equivalence classes of principle G-bundle over a paracompact space B and homotopy classes of maps $[B, BG]$.

Theorem 2.15. *For any topological group G, there exists such a universal bundle. Moreover, a principal G-bundle over a paracompact space is universal if and only if its total space is contractible.*

Proof. There is a wealth of literature on this topic. [BCM, Theorem 1.26] points to various proofs of the result. The version stated here is from [Dol63, Theorem 7.5]. □

Corollary 2.16. *There is a 1-1 correspondence between equivalence classes of principal G-bundles over a fixed base space B and homotopy classes of maps $[B, BG]$.*

Example 2.17. An important example is vector bundle. For a real vector bundle, the structure group is the general linear group $GL_n(\mathbb{R})$. However, note that there is a deformation retraction of $GL_n(\mathbb{R})$ onto the orthogonal group $O(n)$ by the Gram-Schmidt process. Thus, we may consider the structure group to be the orthogonal group. In this case, the classifying space $BO(n)$ is the infinite Grassmannian $G_n(\mathbb{R}^\infty)$: the space of all n-planes in $\mathbb{R}^\infty$. Similarly, for complex vector bundle, the structure group deformation retracts onto the unitary group $U(n)$. The classifying space $BU(n)$ is the infinite complex Grassmannian $G_n(\mathbb{C}^\infty)$ defined similarly to the real case. Studying the cohomology of these classifying space is one of the main themes of studying characteristic classes, which we will discuss in the next section.

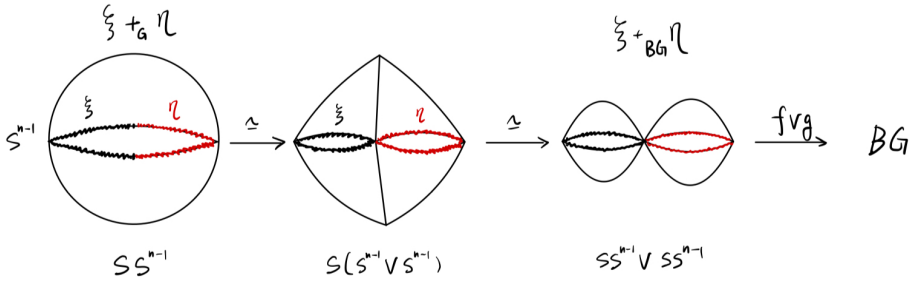
Note that this also applies to the case of bundles over spheres. Therefore, for bundles over S^n with structure group G, we obtain two correspondence: $\pi_{n-1}(G)$ and $\pi_n(BG)$. These two should be the same thing. Indeed, one quick way to verify this is to look at the long exact sequence of homotopy groups associated to the universal bundle $G \to EG \to BG$. The total space EG is contractible, so the homotopy group shifts a dimension.

This long exact sequence interpretation implies that in the special case of bundles over spheres, there should be a stronger relation, namely that the isomorphism between them should preserve the bundle they represent. More precisely, since there is a natural group structure on homotopy groups, this allows us to 'add' two fiber bundles. There are two ways to do this: via identification with $\pi_{n-1}(G)$, or via identification with $\pi_n(BG)$. We will show that these two operations coincide.

Proposition 2.18. *The map $\Phi : \pi_{n-1}(G) \to \pi_n(BG)$ sending $[\xi] \mapsto \langle \xi \rangle$ is an isomorphism of groups, where $[\cdot], \langle \cdot \rangle$ denote the element the fiber bundle ξ represents in $\pi_{n-1}(G)$ and $\pi_n(BG)$ respectively.*

Proof. It suffices to show that this is a group homomorphism. Let ξ and η be two fiber bundles, and ω be the universal bundle. To avoid confusion, we write $\xi +_G \eta$ when they are added in $\pi_{n-1}(G)$, and write $\xi +_{BG} \eta$ when they are added in $\pi_n(BG)$. Let $f, g : S^n \to BG$ such that $f^*\omega = \xi$ and $g^*\omega = \eta$. The proof is essentially captured in the below figure.

The first two horizontal arrows, which is a deformation retraction, can be formally expressed as $SS^{n-1} \simeq S(S^{n-1} \vee S^{n-1}) \simeq SS^{n-1} \vee SS^{n-1}$, where SX denotes the suspension of X. The leftmost



S^n illustrate $\xi +_G \eta$, while the right most $S^n \vee S^n$ illustrate $\xi +_{BG} \eta$ as $(f \vee g)^*\omega$ is just $f^*\omega$ and $g^*\omega$ on the two components of the wedge. The deformation retraction shows that they coincide. □

Finally, we look at the specific case of S^3 bundles over S^4 with structure group $SO(4)$, which is the group of orientation-preserving rotations of S^3. Because the dimension is low enough, we are able to give a complete classification. By theorem 2.14, the equivalence classes of such bundles are in 1-1 correspondence with $\pi_3(SO(4))$. As a topological space, $SO(4)$ is homeomorphic to $S^3 \times \mathbb{R}P^3$, and is doubly-covered by $S^3 \times S^3$. In fact, this covering map can be given explicitly, thanks to the existence of the quaternion algebra $\mathbb{H}$. It is given by

$$\begin{aligned} p : S^3 \times S^3 &\to SO(4) \\ (u_1, u_2) &\mapsto (v \mapsto u_1 v\, u_2) \end{aligned}$$

where quaternion multiplication is understood on the right. This shows that

$$\pi_3(SO(4)) \cong \pi_3(S^3 \times S^3) \cong \pi_3(S^3) \times \pi_3(S^3) \cong \mathbb{Z} \oplus \mathbb{Z}.$$

Again, using the quaternion, there is a very simple way to identify $\pi_3(S^3)$ with $\mathbb{Z}$. Namely, they are powers of quaternions: $z \mapsto z^n$. Combining these results, the identification between $\mathbb{Z} \oplus \mathbb{Z}$ can be expressed as

$$\begin{aligned} f_{hj} : S^3 &\to SO(4) \\ u &\mapsto (v \mapsto u^h v\, u^j) \end{aligned}$$

for any $h, j \in \mathbb{Z}$. Every element in $\pi_3(SO(4))$, or every pair (h, j), determines a unique equivalence class of S^3 bundle over S^4 with structure group $SO(4)$, which we will denote from now on as ξ_{hj}.

By theorem 2.8, the structure of ξ_{hj} can be described explicitly as follows. Take two charts on S^4, which are the one omitting the north pole and the one omitting the south pole. Identify these two charts with $\mathbb{R}^4$ by stereographic projection. A point on S^4 that corresponds to $u \in \mathbb{R}^4$ in one chart will corresponds to $u/||u||^2$ in the other chart. Form the trivial bundle of S^3 over each of them, making them the product space $\mathbb{R}^4 \times S^3$. What remains is to identify the two parts $(\mathbb{R}^4 - \{0\}) \times S^3$, which corresponds to the 'middle' area where the two charts overlap. This identification is

$$(u, v) \mapsto (u', v') = \left(\frac{u}{||u||^2}, \frac{u^h v\, u^j}{||u||^{h+j}}\right),$$

where the norm $||u||$ is added to compensate the fact that we identify $S^4 - \{0\}$ with $\mathbb{R}^4$ via the stereographic projection.

This can be read as: for every base point $x \in S^4$ that is not the north pole or the south pole, identify the two 3-spheres over them by a rotation. If we write x in coordinate as $(x_1, x_2, x_3, x_4, x_5)$, the stereographic projection maps x to $x' = (\frac{x_1}{1-x_5}, \frac{x_2}{1-x_5}, \frac{x_3}{1-x_5}, \frac{x_4}{1-x_5})$. Then the rotation is specified by $f_{hj}(x'/||x'||)$. Notice that this rotation is independent of x_5 (because of the normalization in the end) and only depends on the ratio between x_1, x_2, x_3, x_4. This corresponds to the previous fact that

the equivalence classes of such bundles only depend on homotopy class of maps from $S^3 \times (-\epsilon, \epsilon)$ to $\pi_3(SO(4))$ so we may 'copy' the identification on the equator S^3 to every circle of latitude.

The case of $h, j = 0$ gives the trivial bundle because f_{hj} is the identity map, so the two maps $(\mathbb{R}^4 - \{0\}) \times S^3 \hookrightarrow S^4 \times S^3$ are compatible, and the whole space is homeomorphic to the product space $S^4 \times S^3$.

Another familiar example of S^3 bundle over S^4 is the quaternion Hopf fibration. [Hat22, Example 4.46] Note that, *a priori*, we can't say affirmatively that it belongs to one of the equivalence classes above. This is because the structure group we have chosen is $SO(4)$, which excludes all orientation-reversing automorphism of S^3, which *could* be necessary in building the Hopf fibration. Nevertheless, the Hopf fibration indeed represents an element in $\pi_3(SO(4))$, and we will use a direct calculation to show which one it is.

The quaternion Hopf fibration can be described as follows. Write any point $z \in S^7$ in coordinate $z = (z_1, z_2)$ where z_1, z_2 each contains a tuple of four coordinates. The projection $p : S^7 \to S^4$ is given by $(z_1, z_2) \mapsto z_1 z_2^{-1}$, where z_1, z_2 are seen as quaternions, according to their coordinates. The quaternion $z_1 z_2^{-1}$ lies in $\mathbb{H} \cup \{\infty\}$, which is identified with S^4 via the stereographic projection where $\mathbb{H}$ is identified with $S^4 \backslash \{N\}$, and $\mathbb{H} \cup \{\infty\} \backslash \{0\}$ is identified with $S^4 \backslash \{S\}$. The two charts will be constructed on $U_N := p^{-1}(S^4 \backslash \{N\})$ and $U_S := p^{-1}(S^4 \backslash \{S\})$. Note that U_N contains all the points (z_1, z_2) with $z_2 \neq 0$, and that U_S contains all the points (z_1, z_2) with $z_1 \neq 0$. Their trivialization is given by:

$$\begin{aligned} \phi_N : & U_N \to \mathbb{R}^4 \times S^3 \\ & (z_1, z_2) \mapsto (z_1 z_2^{-1}, \frac{z_2}{|z_2|}) \\ \phi_N : & U_S \to \mathbb{R}^4 \times S^3 \\ & (z_1, z_2) \mapsto (\overline{z_2 z_1^{-1}}, \frac{z_1}{|z_1|}) \end{aligned}$$

Then the transition function is

$$\begin{aligned} \phi_S \, \phi_N^{-1} : \mathbb{R}^4 \times S^3 &\to \mathbb{R}^4 \times S^3 \\ (u, v) &\mapsto (\frac{u}{||u||^2}, \frac{v \, u}{||u||}) \end{aligned}$$

This shows the following fact that we will return to in later arguments.

Proposition 2.19. *The quaternion Hopf fibration corresponds to $\xi_{0,1}$ in the above classification.*

3. Characteristic classes and the Hirzebruch signature theorem

Characteristic classes are invariants of vector bundles that measure how 'twisted' they are. In this section we give an efficient introduction, leading to the Hirzebruch signature theorem and a computation of the first Pontryagin class of the quaternion Hopf fibration. These two results will play an important role in showing that the differential structure we construct is indeed exotic.

Before getting into the discussion of characteristic classes, we first introduce some basic operations and examples of vector bundles that may not be covered in a standard smooth manifold class.

Definition 3.1. Given two vector bundles ξ_1 and ξ_2 with projection maps $p_1 : E_1 \to B_1, p_2 : E_2 \to B_2$. The *Cartesian product bundle* $\xi_1 \times \xi_2$ is defined to be the vector bundle with projection map $p_1 \times p_2 : E_1 \times E_2 \to B_1 \times B_2$.

Definition 3.2. Given two vector bundles ξ_1 and ξ_2 over the same base space B. Let $d : B \to B \times B$ be the diagonal embedding. The bundle $d^*(\xi_1 \times \xi_2)$ is called the *Whitney sum* of ξ_1 and ξ_2, and is denoted $\xi_1 \oplus \xi_2$.

Note that the fiber of each point in $\xi_1 \oplus \xi_2$ is the direct sum of the fiber in ξ_1 and ξ_2. Thus, the Whitney sum is a formal way to 'add' two vector spaces point-wise.

Example 3.3. The *tautological line bundle* γ_n^1 over $\mathbb{R}P^n$ is defined as follows. A point on $\mathbb{R}P^n$ may be viewed as an unordered pair $\{x, -x\}$ where x lies on S^n. The total space $E(\gamma_n^1)$ is defined as a subspace of $\mathbb{R}P^n \times \mathbb{R}^{n+1}$ consisting pairs $(\{\pm x\}, v)$ where v is a multiple of x. Thus each fiber is a line through the origin.

Proposition 3.4. *The tautological line bundle γ_n^1 over $\mathbb{R}P^n$ is not trivial.*

Proof. Every trivial vector bundle possesses a nowhere zero cross-section. We will show that γ_n^1 does not possess one. Let $s : \mathbb{R}P^n \to E(\gamma_n^1)$ be any cross-section. Consider the composition

$$S^n \to \mathbb{R}P^n \to E(\gamma_n^1),$$

which maps $x \mapsto (\{\pm x\}, t(x)x)$ where $t(x)$ is a continuous real-valued function. Then we have $t(x) = -t(-x)$. It immediately follows that $f(x) = 0$ for some x, and that every cross-section is zero at some point. □

Example 3.5. The construction above can also be applied to the case of spheres. However, it is easy to see that such bundle is trivial. This is because there is a nowhere zero section (by choosing the direction outwards of the sphere). Denote this bundle by ε^1. Then the Whitney sum with the tangent bundle $\tau(S^n) \oplus \varepsilon^1$ is isomorphic to ε^{n+1}, the trivial rank $n+1$ bundle. A bundle whose Whitney sum with some trivial bundle is trivial is called *stably trivial*.

Definition 3.6. A vector bundle ξ is oriented if each fiber (as a vector space) is oriented, and these orientations are locally consistent.

The basic idea of characteristic classes is as follows. Recall that equivalence classes of vector bundles corresponds to homotopy classes of maps f from the base space to a classifying space. If we fix a cohomology class α in the classifying space, then the cohomology class $f^*\alpha$ is well-defined. This is, in a sense, the most general definition of characteristic classes: they are pull-backs of certain cohomology classes of the classifying space. However, how much information this class captures depends on the choice of the class we pull back. Also, this definition is a bit inconvenient to work with: the specific bundle map to the classifying can be difficult to analyze. As such, we will define characteristic classes in an axiomatic way. That is, we will assume we can pick certain cohomology classes (in certain coefficients) of the classifying space such that the pull-backs satisfy some good properties.

We first deal with the case of real vector bundles. For real vector bundles, there is a problem of orientation. Therefore, $\mathbb{Z}_2$ coefficient is used.

Theorem-Definition 3.7. There is a unique sequence of functions $w_1, w_2, w_3, ...$ assigning to each real vector bundle ξ a cohomology class $w_i(\xi) \in H^i(B; \mathbb{Z}_2)$, satisfying four axioms:

(1) The class $w_0(\xi)$ is the unit element $1 \in H^0(B; \mathbb{Z}_2)$, and $w_i(\xi)$ is zero for i greater than the rank of the vector bundle.
(2) **Naturality:** $w_i(f^*\xi) = f^*(w_i(\xi))$, where $f^*\xi$ is the pullback bundle.
(3) **Whitney sum formula:** $w(\xi \oplus \eta) = w(\xi) \smile w(\eta)$, where $w = 1 + w_1 + w_2 + ... \in H^\bullet(B, \mathbb{Z}_2)$.
(4) For the tautological line bundle γ_1^1 over $\mathbb{R}P^1$, $w_1(\gamma_1^1)$ is a generator of $H^1(\mathbb{R}P^1, \mathbb{Z}_2)$.

The classes $w_1(\xi), w_2(\xi), ...$ is called the *Stiefel-Whitney classes* of ξ.

Proposition 3.8. *Any trivial or stably trivial vector bundle has all Stiefel-Whitney class trivial. In particular, the Stiefel-Whitney class cannot completely classify real vector bundles.*

Proof. Any trivial bundle has trivial Stiefel-Whitney classes. This is because they are the pull-back bundle of a rank n bundle over a point. By the first and second axiom, the Stiefel-Whitney

class is trivial. If a bundle is stably trivial, then by the Whitney sum formula, its Stiefel-Whitney class is also trivial. In particular, this means that the tangent bundle of spheres always has trivial Stiefel-Whitney class. However, it is an established fact that S^n is parallelizable if and only if $n = 1, 3, 7$. □

For any complex vector bundles, its underlying real vector bundle is canonically oriented (see e.g. [MS74, Lemma 14.1]), thus $\mathbb{Z}$ coefficient is used. The axiomatic definitions for the real and complex cases are analogous.The tautological line bundle is defined similarly in the complex case.

Theorem-Definition 3.9. There is a unique sequence of functions $c_1, c_2, c_3, ...$ assigning to each complex vector bundle ξ a cohomology class $c_i(\xi) \in H^{2i}(B; \mathbb{Z})$, satisfying four axioms:

(1) The class $c_0(\xi)$ is the unit element $1 \in H^0(B; \mathbb{Z})$, and $c_i(\xi)$ is zero for i greater than the rank of the vector bundle.
(2) **Naturality:** $c_i(f^*\xi) = f^*(c_i(\xi))$, where $f^*\xi$ is the pullback bundle.
(3) **Whitney sum formula:** $c(\xi \oplus \eta) = c(\xi) \smile c(\eta)$, where $c = 1 + c_1 + c_2 + ... \in H^\bullet(B, \mathbb{Z})$.
(4) For the tautological line bundle γ_1^1 over $\mathbb{C}P^1$, $c_1(\gamma_1^1)$ is a generator of $H^2(\mathbb{C}P^1, \mathbb{Z})$.

The classes $c_1(\xi), c_2(\xi), ...$ is called the *Chern classes* of ξ.

We mentioned before that these are pull-backs of certain 'nice' cohomology class of the classifying space. The following theorems make this precise.

Theorem 3.10. *Let γ^n denotes the universal bundle over the classifying space $BO(n)$. Then the cohomology ring $H^*(BO(n); \mathbb{Z}_2)$ is a polynomial algebra over $\mathbb{Z}_2$ freely generated by the Stiefel-Whitney classes $w_1(\gamma^n), w_2(\gamma^2), ..., w_n(\gamma^n)$.* [MS74, Theorem 7.1]

Theorem 3.11. *Let γ^n denotes the universal bundle over the classifying space $BU(n)$. Then the cohomology ring $H^*(BU(n); \mathbb{Z})$ is a polynomial algebra over $\mathbb{Z}$ freely generated by the Chern classes $c_1(\gamma^n), c_2(\gamma^2), ..., c_n(\gamma^n)$.*[MS74, Theorem 14.5]

Remark 3.12. To actually show that there exist such characteristic classes, one approach is to simply define them as the pull-backs of the cohomology classes listed in the above two theorems and verify that they satisfy the four axioms. This involves computing the cup product of these cohomology classes in $BO(n)$ and $BU(n)$, which is not too enlightening. Different constructions using completely different approach can be found in [MS74, Ch. 8, 14] and [Hat17, Theorem 3.1, 3.2], which also shed light on other interpretations of characteristic classes, for example, as obstructions.

Using $\mathbb{Z}_2$ coefficient in the real case inevitably makes us lose some information. There are two type of refinements we can make to obtain information in $\mathbb{Z}$ coefficients. They are the *Euler class* and the *Pontryagin class*.

Theorem 3.13. *Let ξ be an oriented rank n vector bundle with total space E. Let E_0 denotes the set of all non-zero elements of E. Then the cohomology group $H^i(E, E_0; \mathbb{Z})$ is zero for $i < n$, and $H^n(E, E_0; \mathbb{Z})$ contains a unique cohomology class u whose restriction $u|(F, F_0)$ is equal to the preferred generator for the orientation on each fiber of ξ. Furthermore, the map $y \mapsto y \smile u$ is an isomorphism between $H^k(E; \mathbb{Z})$ and $H^{k+n}(E, E_0; \mathbb{Z})$ for every integer k.*

Proof. See [MS74, Ch.10]. A de Rham cohomology approach is in [Loo26, Proposition 4.3]. □

The inclusion $(E, \emptyset) \subset (E, E_0)$ induces a homomorphism $H^*(E, E_0; \mathbb{Z}) \to H^*(E; \mathbb{Z})$. The image of u under this homomorphism is denoted $u|E$. Furthermore, the projection $p^* : H^*(B; \mathbb{Z}) \to H^*(E; \mathbb{Z})$ is an isomorphism. This gives to the following definition.

Definition 3.14. The *Euler class* of an oriented rank n vector bundle ξ is the cohomology class $e(\xi) \in H^n(B; \mathbb{Z})$ that equals to $p^{*-1}(u|E)$.

We will need the following result.

Proposition 3.15. *For any oriented rank n vector bundle ξ, there is an exact sequence*

$$\cdots \longrightarrow H^i(B) \xrightarrow{\cup e} H^{i+n}(B) \xrightarrow{\pi_0^*} H^{i+n}(E_0) \longrightarrow H^i(B) \xrightarrow{\cup e} \cdots$$

in integer coefficients. The map $\pi_0 : E_0 \to B$ is the projection. This is called the Gysin sequence.

Proof. Start from the long exact sequence of cohomology of the pair (E, E_0) and switch the terms $H^j(E, E_0)$ by $H^{j-n}(E)$ by Theorem 3.13. See also [MS74, Theorem 12.2]. □

One can see the Euler class as a refinement in the top dimension. The Pontryagin class takes a different approach. The idea is to exploit the fact that any complex vector bundle is canonically oriented. For any real vector bundle ξ, at each fiber, form the complexification $F \otimes_{\mathbb{R}} \mathbb{C}$. This new bundle will be denoted $\xi \otimes \mathbb{C}$. One can show that this bundle is self-dual, meaning that $\xi \otimes \mathbb{C}$ is isomorphic (equivalent) to $\overline{\xi \otimes \mathbb{C}}$. This implies that the odd-dimensional Chern class of $\xi \otimes \mathbb{C}$ are all elements of order 2. [MS74, Lemma 15.1] As such, the Pontryagin class will only be defined on dimensions divisible by 4.

Definition 3.16. For a real vector bundle ξ, the i-th *Pontryagin class* $p_i(\xi) \in H^{4i}(B; \mathbb{Z})$ is defined to be the cohomology class $(-1)^i c_{2i}(\xi \otimes \mathbb{C})$.

There is an analogous result to Theorem 3.10 and 3.11.

Theorem 3.17. *Let Λ be an integral domain containing $\frac{1}{2}$, then the cohomology ring $H^*(BSO(n); \Lambda)$ is generated by the Pontryagin class $p_1, ..., p_{[\frac{n}{2}]}$, and the Euler class e. They are subject to the following relations: when n is odd, $e = 0$; when n is even, $e^2 = p_{n/2}$.* [MS74, Theorem 15.9]

Now we are ready to state the Hirzebruch Signature Theorem. The theorem relates the signature and the Pontryagin numbers of a manifold whose dimension is divisible by 4. We first define these two terms.

Definition 3.18. Let M^{4n} be a smooth, compact, oriented manifold. For each partition $I = i_1, ..., i_r$ of n, the I-th *Pontryagin number* $p_I[M^{4n}]$ is defined to be the integer

$$\langle p_{i_1}(\tau(M^{4n})) \cdots p_{i_r}(\tau(M^{4n})), \mu_{4n} \rangle$$

where μ_{4n} is the fundamental class.

Definition 3.19. Let M^{4n} be a compact, oriented manifold. Let $a \in H^{2n}(M^{4n}; \mathbb{Q})$. Define the *signature* σ to be the index of the quadratic form $a \mapsto \langle a \smile a, \mu \rangle$.

In other words, choose a basis $a_1, ...a_r$ such that the matrix associated to the quadratic form is diagonal. The index is the number of positive entries minus the number of negative entries. This is well-defined by the Sylvester's law of inertia. That this quadratic form is non-degenerate follows from the Poincaré duality [Hat22, Proposition 3.38].

So why there is a relation between these two notions? One of the reason is because they are both cobordism invariants. If two compact manifolds cobound a smooth, compact manifold of one dimension higher, they are called cobordant. The n-dimensional oriented cobordism classes of manifolds form an additive group under disjoint union, denoted as Ω_n. The topological product further makes Ω_* into a graded ring. It is shown that the Pontryagin number p_I and the signature σ both induce homomorphism on $\Omega_* \otimes \mathbb{Q} \to \mathbb{Q}$. The ring $\Omega_* \otimes \mathbb{Q}$ has a very simple description: it is the polynomial algebra with generators represented by $\mathbb{C}P^{2k}$, $k \geq 1$. This is a very small generating set. If, by choosing different partitions I, the various Pontryagin numbers span this homomorphism, then it is clear that the signature can be expressed as a linear combination of Pontryagin numbers with coefficients in $\mathbb{Q}$. This is indeed the case, and a general formula is found by Hirzebruch in 1954.

The general formula takes a bit efforts to state. Since we only need the case when $k = 2$, we only state the low-dimensional cases.

Theorem 3.20 (The Hirzebruch Signature Theorem)**.** *For any oriented, closed, and smooth manifold M^{4k}, its signature $\sigma(M^{4k})$ can be expressed as a linear combination of its Pontryagin numbers with coefficients depending only on k. In particular, if we write $\sigma(M^{4k}) = [L_k(p_1, p_2, ..., p_k), \mu]$, then the low-dimensional cases are:*

$$\begin{aligned} L_1 &= \tfrac{1}{3}p_1, \\ L_2 &= \tfrac{1}{45}(7p_2 - p_1^2), \\ L_3 &= \tfrac{1}{945}(62p_3 - 13p_2p_1 + 2p_1^3). \end{aligned}$$

Proof. See [MS74, Theorem 19.4] where a general formula is given for any k. After the formula is given, it boils down to checking the cases of $\mathbb{C}P^{2k}$. □

Finally, we will compute the first Pontryagin class of the quaternion Hopf fibration, which will be used in later arguments. We may switch the fiber space to $\mathbb{R}^4$ because the structure group $SO(4)$ still acts on it effectively. We will compute the Pontryagin by starting from the Euler class, and use the relation between the Euler class, the Chern class, and the Pontryagin class to obtain the result.

Proposition 3.21. *Let ι be a standard generator of $H^4(S^4)$. The first Pontryagin class $p_1(\xi_{0,1})$ of the quaternion Hopf fibration is equal to $\pm 2\iota$.*

Proof. By previous result, we denote the quaternion Hopf fibration by $\xi_{0,1}$. By the Gysin sequence when $i = 0$, there is an exact sequence

$$\cdots \longrightarrow H^3(E_0) \longrightarrow H^0(B) \xrightarrow{\cup e} H^4(B) \xrightarrow{\pi_0^*} H^4(E_0) \longrightarrow \cdots$$

Note that the non-zero part of the total space deformation retracts onto the original total space of the sphere bundle. That is, $E_0 \simeq E(\xi_{0,1}) \simeq S^7$. Thus $\cup e : H^0(B) \to H^4(B)$ is an isomorphism. This shows that the Euler class $e(\xi_{0,1})$ is equal to $\pm\iota$.

Next note that the quaternion Hopf fibration actually admits a complex structure. A complex structure is a continuous map $J : E \to E$ which maps each fiber $\mathbb{R}$-linearly into itself, and satisfies $J(J(v)) = -v$ for any vector v in the fiber. The map we choose here is $v \mapsto i \cdot v$ when written in coordinates as quaternions. Recall that the transition function on the $\xi_{0,1}$ is given by

$$\begin{aligned} \phi_S\, \phi_N^{-1} : \mathbb{R}^4 \times S^3 &\to \mathbb{R}^4 \times S^3 \\ (u, v) &\mapsto \left(\frac{u}{||u||^2}, \frac{v\, u}{||u||}\right) \end{aligned}$$

where the multiplication is on the right. This shows that this map is invariant under the transition function and is well-defined.

Now that we have shown that $\xi_{0,1}$ can be viewed as a complex vector bundle. We denote this as $\xi_{\mathbb{C}}$. The top Chern class $c_2(\xi_{\mathbb{C}})$ is equal to the Euler class. [MS74, Definition 14.1] So $c_2(\xi_{\mathbb{C}}) = \pm\iota$. Finally, by the relation between the Chern class and the Pontryagin class for a complex vector bundle [MS74, Corollary 15.5],

$$p_1(\xi_{0,1}) = c_1(\xi_{\mathbb{C}})^2 - 2c_2(\xi_{\mathbb{C}}) = \pm 2\iota.$$

This completes the proof. □

4. Construction of the exotic sphere I

We begin the construction of the exotic 7-sphere in this section. This will be divided into two parts: this section deals with diffeomorphism while the next section deals with homeomorphism. The exotic nature of the problem makes it clear that we will approach these two problems in two

distinct ways as no standard tool can detect homeomorphism and diffeomorphism separately. This also means that the result in these two sections are entirely independent, and in this section we may sometimes assume that we already know that the space is homeomorphic to the 7-sphere.

The spaces to be considered are the total space of S^3 bundles over S^4 with structure group $SO(4)$. As per our discussion in section 2, they correspond to elements in $\mathbb{Z} \oplus \mathbb{Z}$. The bundles are denoted ξ_{hj}, and the total space of ξ_{hj} is denoted M_{hj}. There is a natural differential structure on it based on the fiber bundle structure.

Proposition 4.1. *For any fiber bundle $F \to E \to B$, if the fiber space F and the base space B both admits a differential structure, and that the structure group G acts on the fiber space by diffeomorphism, then there is a natural differential structure on the total space E.*

Proof. By definition, the base space B has an open cover $\{V_j\}_{j\in J}$ such that for any $j \in J$, there is a homeomorphism $\phi_j : V_j \times F \to p^{-1}(V_j)$. Each $V_j \times F$ admits a differential structure and we declare ϕ_j to be a diffeomorphism. This makes each $p^{-1}(V_j)$ a differential manifold. To show that these 'local' differential structures are compatible, it suffices to show that the identity map $p^{-1}(V_i \cap V_j) \to p^{-1}(V_i \cap V_j)$ is a diffeomorphism where different differential structures are understood on each side. Then it suffices to show that the map

$$\phi_j^{-1}\phi_i : V_i \cap V_j \times F \to V_i \cap V_j \times F$$

is a diffeomorphism. This is clear as it is the identity on $V_i \cap V_j$ and for each fixed $x \in V_i \cap V_j$, the map on F is a diffeomorphism by our assumption. □

This will be the differential structure we put on M_{hj}. There is another way to describe it. Associated with each 3-sphere bundle ξ_{hj} there is a 4-disk bundle ζ_{hj}. Its total space is denoted B_{hj}. The differential structure on B_{hj} can be obtained in the same way. Then the differential structure on M_{hj} is obtained by considering M_{hj} as the smooth boundary of B_{hj}. One can check that these two differential structure coincide.

The idea of showing that certain M_{hj}, with the differential structure described above, is not diffeomorphic to the standard 7-sphere is to use the Hirzebruch Signature Theorem, where a numerical relation is given between the index and the Pontryagin numbers. While the index of a manifold can be defined purely topologically, the Pontryagin numbers depend on the structure of the tangent bundle. Since h, j can take any integer, there is some choices we can make. It is therefore hoped that certain choices of h, j will produce peculiar numbers in the formula and show that the differential structure differs from the standard one.

Note, however, that the signature theorem only applies to closed manifolds. Therefore, it is necessary to consider a 'relative' version. This involves an invariant λ we define below.

Let M^7 be any closed, oriented 7-manifold that is the smooth boundary of an 8-manifold B^8. Further assume that $H^3(M^7) = H^4(M^7) = 0$. Let μ be the fundamental class of M^7. There is a distinguished fundamental class $\nu \in H^8(B^8, M^7)$ determined by $\partial\nu = \mu$. [Hat22, Theorem 3.43] Define the quadratic form over $H^4(B^8, M^7)/(\text{torsion})$ by the formula $\alpha \mapsto \langle \nu, \alpha^2 \rangle$. Let $\tau(B^8)$ denotes the index of this quadratic form. This serves as the 'relative' index. Let $p_1 \in H^4(B^8)$ be the first Pontryagin class of the tangent bundle of B^8. The assumption that $H^3(M^7) = H^4(M^7) = 0$ implies that there is an isomorphism $i^* : H^4(B^8, M^7) \to H^4(B^8)$ and we may 'pull back' the Pontryagin class to $H^4(B^8, M^7)$. The 'relative' Pontryagin number is defined as $q(B^8) = \langle \nu, (i^{*-1}p_1)^2 \rangle$. These two relative notions lead to an invariant.

Theorem-Definition 4.2. The invariant $\lambda(M^7)$ is defined as $2q(B^8) - \tau(B^8) \bmod 7$. This does not depend on the choice of the manifold B^8.

Proof. Let B_1^8 and B_2^8 be any such two manifolds. Let $C^8 = B_1^8 \cup B_2^8$ be the closed 8-manifold where B_1^8 and B_2^8 are glued along the boundary. Choose an orientation ν for C^8 which is consistent

with the orientation ν_1 of B_1^8 and is the reverse of the orientation ν_2 of B_2^8. Let $q(C^8)$ denotes the Pontryagin number $\langle \nu, p_1^2(C^8)\rangle$. Then the theorem follows from the following lemma. □

Lemma 4.3. *Under the above conditions, we have*

$$\tau(C^8) = \tau(B_1^8) - \tau(B_2^8)$$
$$q(C^8) = q(B_1^8) - q(B_2^8)$$

Proof. Consider the following diagram:

$$\begin{array}{ccc} H^4(B_1, M) \oplus H^4(B_2, M) & \xleftarrow{h} & H^4(C, M) \\ \downarrow{\scriptstyle i_1 \oplus i_2} & & \downarrow{\scriptstyle j} \\ H^4(B_1) \oplus H^4(B_2) & \xleftarrow{k} & H^4(C) \end{array}$$

By our assumptions about M, these arrows are all isomorphisms. If $\alpha = jh^{-1}(\alpha_1 \oplus \alpha_2)$, then

$$\langle \nu, \alpha\rangle = \langle \nu, jh^{-1}(\alpha_1^2 \oplus \alpha_2^2)\rangle = \langle \nu_1 \oplus (-\nu_2), \alpha_1^2 \oplus \alpha_2^2\rangle = \langle \nu_1, \alpha_1^2\rangle - \langle \nu_2, \alpha_2^2\rangle.$$

This deals with the index. For the Pontryagin number, note that $k(p_1(C)) = p_1(B_1) + p_1(B_2)$ (this is almost by definition for any characteristic class). Let $\alpha_1 = i_1^{-1}p_1(B_1), \alpha_2 = i_2^{-1}p_1(B_2)$. Then

$$\langle \nu, p_1^2(C)\rangle = \langle \nu_1 \oplus (-\nu_2), \alpha_1^2 \oplus \alpha_2^2\rangle = \langle \nu_1, \alpha_1^2\rangle - \langle \nu_2, \alpha_2^2\rangle.$$

This completes the proof. □

In the next section we will show that when $h + j = 1$, M_{hj} is homeomorphic to S^7. Then the invariant is defined for M_{hj}. To say that the differential structure on M_{hj} is the same as the standard differential structure is to say that M_{hj} is the smooth boundary of the 8-disk. In this case, $\lambda(M_{hj})$ is clearly 0, since the 8-disk is contractible. Thus, if we can manage to compute the invariant λ and show that it is not 0 for some choices of h, j, this shows that these spaces are not diffeomorphic to the standard 7-sphere. This will be our task below.

Since we only show that M_{hj} is homeomorphic to S^7 when $h + j = 1$, these are the cases we will focus on. Let k be any odd integer. Let M_k^7 denoted the total space of ξ_{hj} determined by $h + j = 1, h - j = k$. The computation will be carried out on M_k^7 as the smooth boundary of B_k^8.

Proposition 4.4. *By a choice of orientation, the index $\tau(B_k^8)$ is 1.*

Proof. We have seen that there is an isomorphism $i^* : H^4(B_k^8, M_k^7) \to H^4(B_k^8)$. Further note that B_k^8 deformation retracts onto the base space S^4. Let $p_k : B_k^8 \to S^4$ be the projection map. Then $p_k^* : H^4(S^4) \to H^4(B_k^8)$ is an isomorphism. Now it's clear that $H^4(B_k^8, M_k^7) \cong H^4(S^4) \cong \mathbb{Z}$ so the matrix associated with the quadratic form is 1×1 and the index is just ± 1, depending on the choice of orientation of B_k^8. For future reference, we fix an orientation such that this index is $+1$, and write down the explicit maps. Let ι be a standard generator of $H^4(S^4)$. Let $\alpha = p_k^*(\iota)$, and let $\beta = i^{*-1}(\alpha)$. Now choose fundamental classes μ, ν for M_k^7 and (B_k^8, M_k^7) such that $\langle \nu, \beta^2\rangle = +1$. □

The computation of $q(B_k^8)$, which boils down to the computation of the first Pontryagin class of the tangent bundle of B_k^8, is much more involved. In general, Pontryagin classes are difficult to compute explicitly. We will compute it through a series of lemmas. The first observation is that if we switch the fiber space of ξ_{hj} from S^3 to $\mathbb{R}^4$, it allows for the following decomposition.

Proposition 4.5. *Let $\xi : E \to B$ be a vector bundle. If B is a smooth manifold, then*

$$\tau(E) \cong p^*(\tau(B)) \oplus p^*(\xi),$$

where τ denotes the tangent bundle. Note that E is automatically a smooth manifold when B is.

Proof. Let B be a d-dimensional manifold, and ξ be a rank n vector bundle. For every $x \in B$, there is a small open neighborhood O containing x such that $p^{-1}(O) \simeq \mathbb{R}^d \times \mathbb{R}^n$. Then on this small neighborhood, the total space of the tangent bundle of B can be decomposed as

$$\mathrm{T}E|_{p^{-1}(O)} \simeq \mathrm{T}B|_O \times \mathrm{T}\mathbb{R}^n$$

This decomposition is independent of which chart we choose: we can always tell which direction is 'along' the manifold B, and which direction is 'normal' to B. All is left is to construct a bundle map from E to $B \times B$ according to the above decomposition. □

We have seen in example 3.5 that $\tau(S^n)$ is stably trivial, and thus has trivial Pontryagin class. Now the problem is reduced to computing the Pontryagin class of ξ_{hj}. We abuse the notation and still use ξ_{hj} to denote the bundle when the fiber space is switched to $\mathbb{R}^4$.

Proposition 4.6. *The first Pontryagin class $p_1(\xi_{hj})$ is equal to $\pm 2(h-j)\iota$.*

In order to show this, we will need two lemmas.

Lemma 4.7. *The first Pontryagin class $p_1(\xi_{hj})$ is a linear function on h and j.*

Proof. The main technical part was dealt with in proposition 2.18. The idea is to pull back Pontryagin class of the classifying space to obtain a homomorphism. More specifically, we have

$$\pi_3(SO(4)) \xrightarrow{\Phi} \pi_4(BSO(4)) \xrightarrow{\Theta} H^4(S^4),$$

where $\Theta : f \mapsto f^*(p_1(BSO(4)))$. By proposition 2.18, this is a homomorphism. Thus, an addition in (h, j) results in an addition in the Pontryagin class. □

Lemma 4.8. *There is an orientation-reversing bundle map between ξ_{hj} and ξ_{-j-h}. In particular, ξ_{hj} is isomorphic to ξ_{-j-h} as vector bundles.*

Proof. The map is given by mapping $z \mapsto z^{-1}$ in each fiber space S^3 (again as quaternions). In ξ_{hj}, every fiber space S^3 over a point that is not the north or the south pole is contained in two charts, and the transition function on every fiber space is $v \mapsto u^h v\, u^j$. If we take the inverse of each element on both charts, this transition function is switched to $v \mapsto (u^h v^{-1} u^j)^{-1} = u^{-j} v\, u^{-h}$. This completes the proof. □

Note that this is not a contradiction to the classification of bundles over spheres because we choose the structure group to be orientation-preserving rotations, while this isomorphism requires orientation-reversing ones. Now we are ready to prove proposition 4.6.

Proof. By lemma 4.7, $p_1(\xi_{hj})$ is in the form $(ah + bj)\iota$ for some a, b. Furthermore, Pontryagin class is independent of orientation. Thus, by lemma 4.8, $p_1(\xi_{hj})$ should be the same as $p_1(\xi_{-j-h})$. This implies that $p_1(\xi_{hj})$ is in the form of $a(h-j)\iota$. Finally, by proposition 3.21, the first Pontryagin class of the quaternion Hopf fibration is $\pm 2\iota$. Since we know the quaternion Hopf fibration corresponds to the case $h = 0, j = 1$, this shows that $a = \pm 2$. □

Theorem 4.9. *The invariant $\lambda(M_k^7)$ is equal to $k^2 - 1 \bmod 7$.*

Proof. By proposition 4.5 and the Whitney sum formula,

$$p_1(B_k^8) = p_1(p^*(\tau(S^4))) + p_1(p^*(\xi_k)) = 0 + p^*(p_1(\xi_k)) = \pm 2k\alpha.$$

Now we have

$$q(B_k^8) = \langle \nu, (i^{*-1}(\pm 2k\alpha))^2 \rangle = \langle \nu, 4k^2\beta \rangle = 4k^2.$$

Thus, $\lambda(M_k^7) = 2q(B_k^8) - \tau(B_k^8) \equiv k^2 - 1 \bmod 7$. This completes the proof. □

We have shown that when k^2-1 is not congruent to 0 mod 7, M_k^7 cannot be diffeomorphic to the standard 7-sphere. Next section deals with the final task of showing that they are homeomorphic to the 7-sphere.

5. Construction of the exotic sphere II

We will show the following result in this section.

Theorem 5.1. *The manifold M_k^7 is homeomorphic to S^7 for any odd integer k.*

The tool to tackle this problem is Morse theory, as any algebraic topological tool cannot distinguish a space up to homeomorphism. While the argument is very straightforward, it is difficult to give a strong motivation. As such, we only derive the necessary ingredients for the proof and do not discuss further.

Definition 5.2. Let f be a smooth real-valued function on a smooth manifold M. A point $p \in M$ is called a *critical point* of f if $\frac{\partial f}{\partial x^i}(p)$ are all 0 in a local coordinate system. A critical point of f is called *non-degenerate* if the *Hessian* matrix

$$\left(\frac{\partial^2 f}{\partial x^i \partial x^j}(p)\right)$$

is non-singular. [Mil63, P4]

Theorem 5.3 (Reeb's Theorem)**.** *If M is a compact manifold and f is a smooth real-valued function on M with only two critical points, both non-degenerate, then M is homeomorphic to a sphere.*

Proof. See [Mil63, Theorem 4.1]. □

Now we are ready to prove the result. By the above theorem, the proof boils down to choose such a suited function on M_k^7.

Proof. The map $f : M_k^7 \to \mathbb{R}$ is defined on the two charts of the bundle. Recall that the two charts, each homeomorphic to $\mathbb{R}^4 \times S^3$, are identified on $(\mathbb{R}^4 - \{0\}) \times S^3$ by the formula

$$(u, v) \mapsto (u', v') = \left(\frac{u}{||u||^2}, \frac{u^h v\, u^j}{||u||}\right).$$

Replace the coordinates (u', v') by (u'', v') where $u'' = u'(v')^{-1}$. The function f is defined as

$$f(x) = \frac{\mathrm{Re}(v)}{(1+||u||^2)^{\frac{1}{2}}} = \frac{\mathrm{Re}(u'')}{(1+||u''||^2)^{\frac{1}{2}}}$$

where $\mathrm{Re}(\cdot)$ stands for the real part of a quaternion. The function is clearly smooth. We first verify that it is well-defined where the two charts overlap. Note that for any two quaternions u, v, the real parts $\mathrm{Re}(uv) = \mathrm{Re}(vu)$ as all the non-commutativity happens in imaginary parts. Thus,

$$\mathrm{Re}(u'') = \frac{1}{||u||}\mathrm{Re}\left(\frac{u}{u^h v\, u^j}\right) = \frac{1}{||u||}\mathrm{Re}(u^{-j} v\, u^{-h+1}) = \frac{1}{||u||}\mathrm{Re}(v).$$

Then,

$$\frac{\mathrm{Re}(u'')}{(1+||u''||^2)^{\frac{1}{2}}} = \frac{\mathrm{Re}(v)}{||u||(1+||u||^{-2}||v||^{-2})^{\frac{1}{2}}} = \frac{\mathrm{Re}(v)}{(1+||u||^2)^{\frac{1}{2}}}.$$

Next we evaluate its critical points. Write $u'' = a + bi + cj + dk$, then

$$f(x) = \frac{a}{(1+a^2+b^2+c^2+d^2)^{\frac{1}{2}}},$$
$$\frac{\partial f}{\partial a} = \frac{1+b^2+c^2+d^2}{(1+||u''||^2)^{\frac{3}{2}}} > 0.$$

Thus only the north pole and south pole can be critical points. Since M_k^7 is compact, they must be the minimum and maximum points, and thus are critical points. We omit the rest of the verification that they are non-degenerate since it's a straightforward but somewhat tedious computation of second derivatives in local coordinates. □

Combining the results in these two sections, we have reached the following conclusion.

Theorem 5.4. *When $k^2 - 1$ mod 7 is not equal to 0, the manifold M_k^7 is homeomorphic but not diffeomorphic to S^7.*

6. Concluding remarks

It is a fascinating journey to construct and show that the 7-sphere possesses different smooth structures. The arguments are very clean and apply tools and knowledge from many different areas. This fascination is continued when in 1963, Milnor and Kervaire showed that there are precisely 28 distinct smooth structure on the 7-sphere, using what's called *h-cobordism theory*. It relates the problem to the stable homotopy group of spheres. See [KM63] for more details.

Acknowledgments

I would like to express my gratitude to Prof. Akhil Mathew for providing guidance on this project and answering my questions during weekly meetings. This project would not be as complete without his generous support. I would also like to thank Prof. Eduard Looijenga for giving us this great opportunity to explore a topic of our interests.

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